

Problem A. Chimpanzini Bananini

Time Limit 2000 ms

Mem Limit 262144 kB

Chimpanzini Bananini stands on the brink of a momentous battle—one destined to bring finality.

For an arbitrary array b of length m , let's denote the *rizziness* of the array to be $\sum_{i=1}^m b_i \cdot i = b_1 \cdot 1 + b_2 \cdot 2 + b_3 \cdot 3 + \dots + b_m \cdot m$.

Chimpanzini Bananini gifts you an empty array. There are three types of operations you can perform on it.

1. Perform a cyclic shift on the array. That is, the array $[a_1, a_2, \dots, a_n]$ becomes $[a_n, a_1, a_2, \dots, a_{n-1}]$.
2. Reverse the entire array. That is, the array $[a_1, a_2, \dots, a_n]$ becomes $[a_n, a_{n-1}, \dots, a_1]$.
3. Append an element to the end of the array. The array $[a_1, a_2, \dots, a_n]$ becomes $[a_1, a_2, \dots, a_n, k]$ after appending k to the end of the array.

After each operation, you are interested in calculating the *rizziness* of your array.

Note that all operations are **persistent**. This means that each operation modifies the array, and subsequent operations should be applied to the current state of the array after the previous operations.

Input

The first line contains an integer t ($1 \leq t \leq 10^4$) — the number of test cases.

The first line of each test case contains an integer q ($1 \leq q \leq 2 \cdot 10^5$) — the number of operations you perform on your array.

The following q lines first contain a single integer s ($1 \leq s \leq 3$) — the operation type.

- If $s = 1$, then the cyclic shift operation should be performed.
- If $s = 2$, then the reversal operation should be performed.
- If $s = 3$, then the line will contain an additional integer k ($1 \leq k \leq 10^6$), denoting the element appended to the back of the array.

It is guaranteed that the sum of q will not exceed $2 \cdot 10^5$ over all test cases. Additionally, it is guaranteed that the first operation on each test case will be one with $s = 3$.

Output

For each test case, output q lines, outputting the *rizziness* of your array after each operation.

Examples

Input	Output
1	1
13	5
3 1	14
3 2	11
3 3	27
1	23
3 4	48
2	38
3 5	74
1	73
3 6	122
2	102
3 7	88
2	
1	

Note

The first six states of the array:

- [1]
- [1, 2]
- [1, 2, 3]
- [3, 1, 2]
- [3, 1, 2, 4]
- [4, 2, 1, 3]